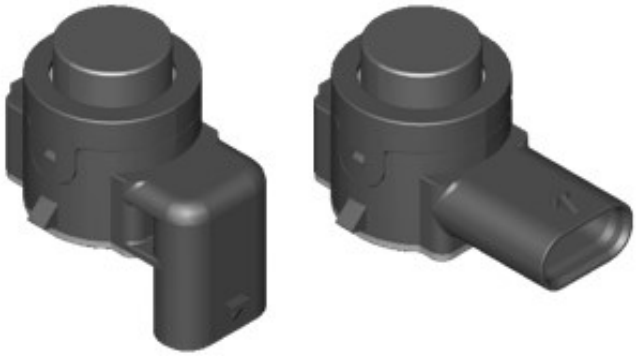
Mounting Specification

Ultrasound Base Development



Copyright by Continental Corporation

All rights reserved

The reproduction, distribution and utilization of this document as well as the communication of its contents to others without express authorization are prohibited. Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or design.

General Information

The abbreviations, the definitions and further applicable documents used for creating the specification are referenced below.

Abbreviations

Abbreviation	Definition
CAD	Computer Aided Design
ISO	International Organization for Standardization
OEM	Original Equipment Manufacturer
TBD	To de defined/determined

Definitions

Term	Definition

Copyright © by Continental Corporation
All rights reserved

Contents

1 Introduction 4

2 Design Mounting Requirements 8
2.1 Pitch Angle β versus Mounting Height h_z 10
2.2 Mounting Tolerances 12

3 Situation of installation 13
3.1 Test coverage 13
3.2 General Guidelines for Bumper Shape 13
3.2.1 Flush Mounting 13
3.2.2 Special Consideration to the placement of sensors next to License Plate 14
3.2.3 Special Consideration to the placement of sensors next to Protuberances 15
3.2.4 Avoidance of sensor placement within or directly next to air turbulence generation systems 15
3.2.5 Special considerations for metal bumper 15
3.3 Sensor Painting 16

4 Decoupling Ring 16

5 Bracket 16

6 Heat Dissipation 17

7 Weight 17

8 Connector Orientation 18

9 Mounting Optimization 18
9.1 Simulation 18
9.1.1 Solution Concept 18
9.1.1.1 First Stage: Optimization 19
9.1.1.2 Second Stage: Verification 21
9.1.1.3 Third Stage: Validation 21
9.1.2 Optimization Setup & Workflow 22
9.1.2.1 Initial sensor setup 22
9.1.2.2 Bumper surface description 23
9.1.2.3 Restrictions 23
9.1.2.4 Antenna templates 24
9.2 Vehicle FoV Test (ISO 17386) 25
9.3 Ground Immunity Adaption 25
9.4 Data capturing based on test driving 25

11 References 26

12 Table of Figure 26

Copyright © by Continental Corporation
All rights reserved

1 Introduction

The US (ultrasonic sensor) is a 52kHz ultrasonic sensor for low speed maneuvering applications. This document is meant to provide a general guideline for sensor mounting and integration into the subject vehicle but makes no claims of being complete.

Continental shall confirm suitable performance of the overall system (ultrasonic sensors integrated in the target platform) and thus confirm final mounting position. The final mechanical drawings and dimensions shall be presented in a separate document.

The ultrasonic sensor is available either with a radial (90°) or an axial (0°) plug that is shown in the following figure:

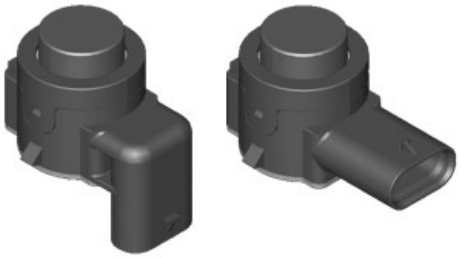


Figure 1: Ultrasonic sensor with a 0° connector (left) and a 90° connector (right)

The sensors are used both for park distance control applications and for automated parking applications, including the parking slot search.

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

An approximation of the field of view of a single sensor is shown in Figure 2 for a flush mounted sensor. In case the sensor is placed within a bezel, the field of view could be different. Ranges for the detection of objects are not in scope of this document and are therefore not inserted into the sensor field of view drawing. In the near range the sensor has a vertical/horizontal FoV angle of 60° and 120°, whereas in the far range the sensor has a reduced FoV angle due to the reached sensitivity limit (+/-6dB) of the ultrasonic beam.

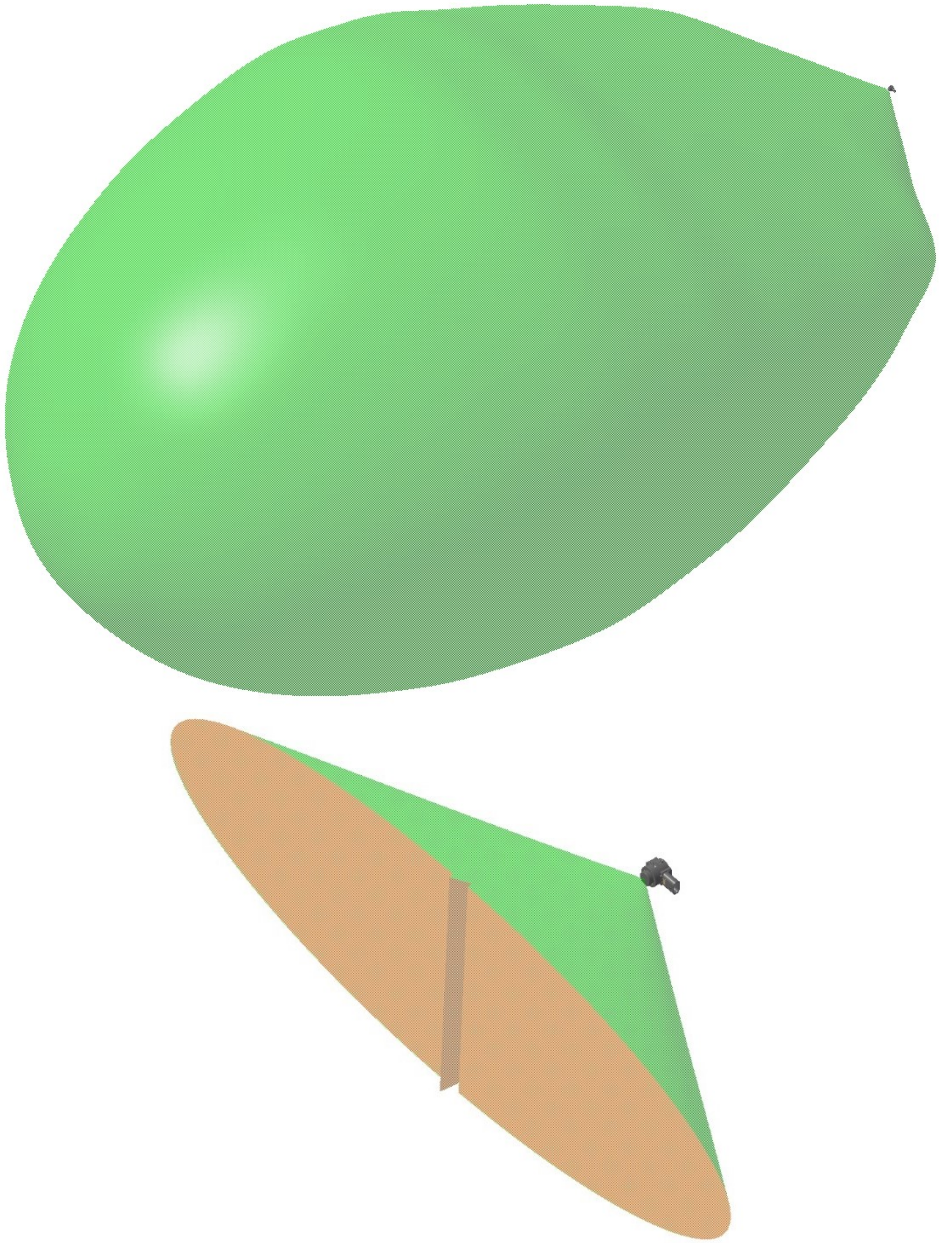


Figure 2: Field of view (FoV) of a single sensor

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

The sensor setup is highly dependent on the vehicle function that needs to be supported with ultrasonic. The following document concentrates on a default 12 sensor setup used for automated parking applications.

The detection field of view of an ultrasonic sensor system is shown exemplary for this 12-sensor setup in the following picture. The detection field of view is the area around the vehicle where an obstacle is detected, but the localization accuracy can vary. The shown parameters in Figure 3 are application dependent and shall be agreed during development between Continental and the customer.

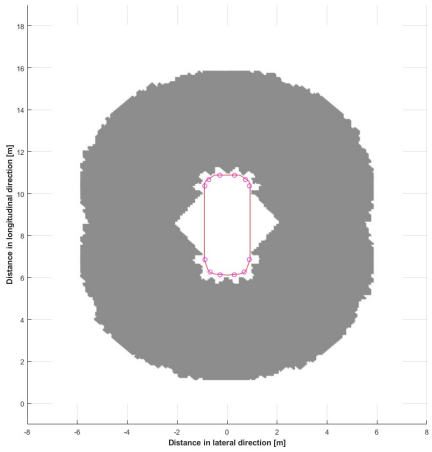


Figure 3: Detection Field of view (FOV) of a 12-sensor setup

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

In the following the ultrasonic sensors are described with different numbers that refer to the mounting position in the vehicle as shown in Figure 4: Sensor naming.



Figure 4: Sensor naming and position

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

2 Design Mounting Requirements

Each sensor can be placed on each mounting position at the bumper, connector facing to the left or to the right for the front/rear sensors and connector facing to the front/rear for the side sensors relative to vehicle x-axis.

α (alpha), β (beta) and γ (gamma) angles are defined in Figure 7: Definition of angles. Sense of rotation of angles (sign) is defined according to the right-hand rule.

Δxy is defined as 0mm if the sensor is flush mounted with the bumper surface. If Δxy is positive, the sensor is moved outside the bumper. If Δxy is negative, the sensor is moved inside the bumper.

	Selection of Mounting position	Assembly Tolerance	
	Nominal values and its mounting windows	**Min**	**Max**
α (with respect to the vehicle x-axis)	Nominal position is dependent on the car geometry and the desired parking functions	-2°	+2°
β	Nominal position is dependent on the car geometry and the desired parking functions	-1°	+1°
γ	0°, 180°	-2°	+2°
Δxy	0mm (flush mounting)	-1mm	+ 0.5mm
h_z	Nominal position is dependent on the car geometry and the desired parking functions	-1.5mm	+1.5mm
y	Needs to be determined once a final car geometry is available. The aim should be for a vehicle field of view with minimal areas of low detectability (“blind spots”). The side sensors US1, 6, 7 and 12 should ideally face the positive and negative y directions. The sensor positions are lying symmetrically on both sides of the xz- plane.	-1.5mm	+1.5mm

Table 1: mounting orientation requirements

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

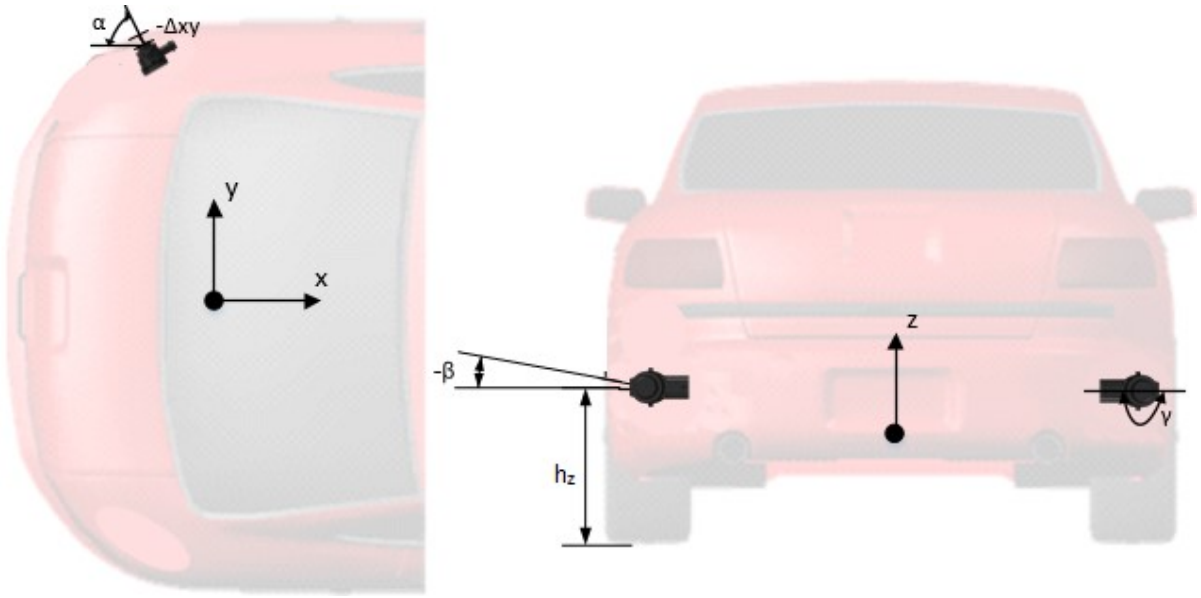


Figure 5: Mounting position and mounting angles

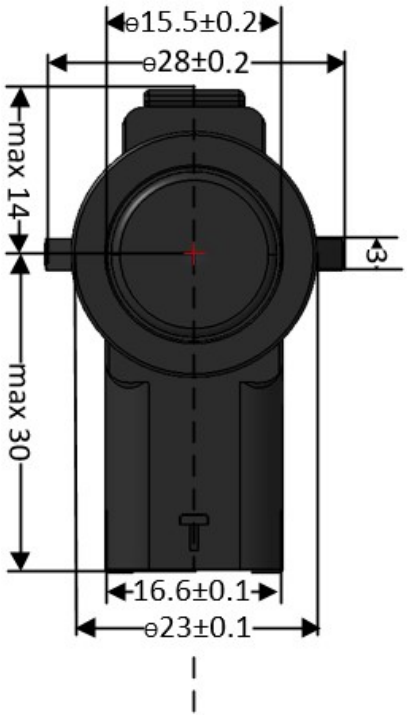


Figure 6: Sensor reference point (center of transducer) for α (alpha), β(beta), γ(gamma), Δy, h_z (Dimensions in mm, final dimensions part of Mechanical drawings)

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

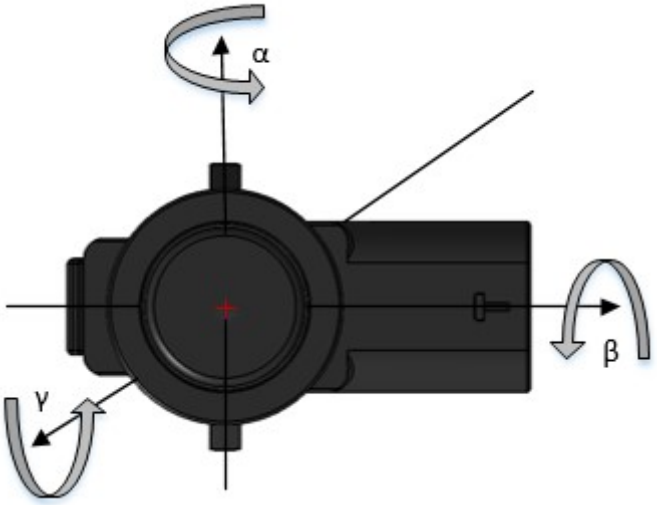


Figure 7: Definition of angles: α, β, γ=0°

2.1 Pitch Angle β versus Mounting Height h_z

The pitch angle β and the mounting height h_z is dependent on each other. The mounting windows are different for the side sensors (US1, US6, US7, US12) and for the front/rear sensors (US2, US3, US4, US5, US8, US9, US10, US11) and shown in Figure 8 and Figure 9. If the sensor is mounted in the bumper with a pitch and height within the green mounting windows of below figures, the sensor performance is ensured. If the sensors are mounted with parameters that lie into the red window, the sensor performance could be affected in a critical way. If a placement of the sensors with this critical parameter cannot be avoided due to customer vehicle application constraints, this must be communicated to Continental in an early stage of development for further feasibility analysis.

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

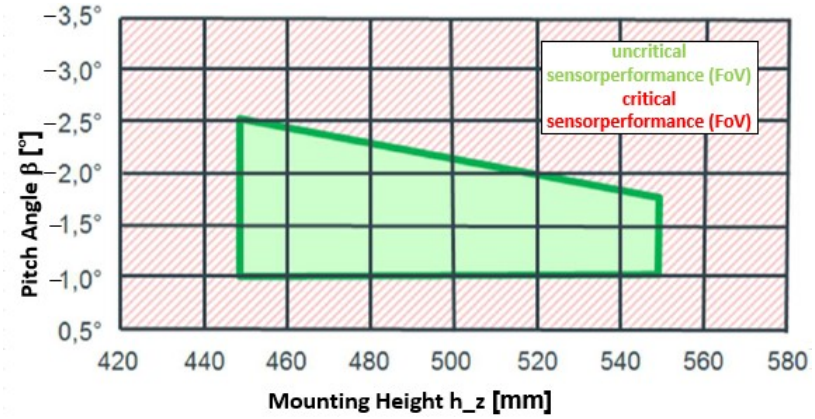


Figure 8: Pitch Angle β versus Mounting Height h_z for side sensors (US1, US6, US7, US12)

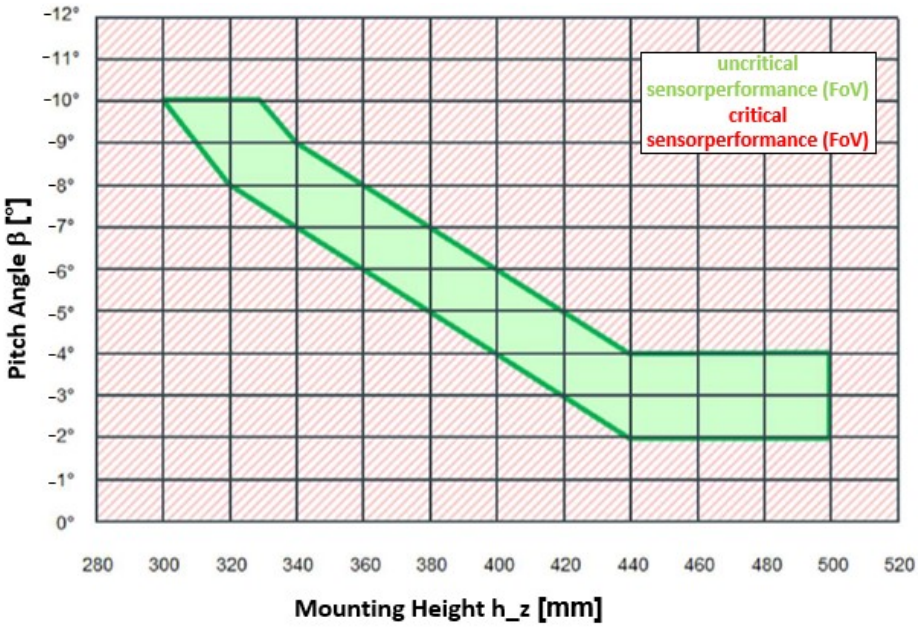


Figure 9: Pitch Angle β versus Mounting Height h_z for front/rear sensors (US2, US3, US4, US5, US8, US9, US10, US11)

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

2.2 Mounting Tolerances

The **azimuthal** mounting tolerance Δα shall not exceed the **value given in Table 1**. In this overall tolerance the following contributions are included:

1: Vehicle Body-Bracket

2: Bracket-Sensor

The mechanical mounting tolerance for the **pitch** angle $\Delta\beta$ shall not exceed the **value given in Table 1**. In this overall tolerance the following contributions are included:

1**:** Vehicle Body-Bracket

2**:** Bracket-Sensor

The mechanical mounting tolerance in **elevation** (roll angle) $\Delta\gamma$ shall not exceed the **value given in Table 1**. In this overall tolerance the following contributions are included:

1**:** Vehicle Body-Bracket

2**:** Bracket-Sensor

Attention must be paid to the facts that:

If not otherwise agreed between Continental and the customer, the given mounting tolerances do not consider additional misalignments imposed by load changes of the subject vehicle. I.e. Additional tolerances imposed by load of passengers and/or luggage etc.

I.e. All functions will be verified to work within

the mounting limits described in this document and

as agreed between Continental and the customer

Additional misalignment, especially in elevation and pitch (by load changes) cannot be compensated by any sensor means.

The mounting window and mounting tolerances given in this document have been determined theoretically and have to be verified by tests

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

3 Situation of installation

In general terms, for a proper sensor performance the hemisphere in front of the sensor needs to be kept free of any materials or objects that may disturb the wave emission characteristics. The environment in the back hemisphere of the sensor needs to be checked together with the customer.

As this cannot be achieved under all circumstances, below sections provide a guidance to minimize impact of vehicle parts in the hemisphere of the sensor.

3.1 Test coverage

To limit the impact of the situation of installation to the system performance, verification tests shall be conducted as early as possible. Test planning and scheduling shall be agreed with Continental at the beginning of a project (car platform development).

To keep test effort on a reasonable level, tests shall be planned and conducted only, when test results can be applied to the final platform.

Appropriate tests are defined within most of the following sections.

3.2 General Guidelines for Bumper Shape

Following guidelines shall be followed for bumper design:

3.2.1 Flush Mounting

The sensor with its decoupling ring shall be flush mounted at the bumper with the tolerances given in table 1.

It needs to be communicated to Continental at an early stage of development if the customer has design constraints that prohibit the flush mounting of the sensors. Non-flush mounted concepts must be evaluated in detail for feasibility with the specific vehicle application parameter. For example, a mounting of the sensors in a recessed way could strongly affect the ultrasonic beam as the recessed mounting forms a vessel around the field of view of the sensor.

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

3.2.2 Special Consideration to the placement of sensors next to License Plate

A canyoning effect happens if the license plate is placed close to the sensor with canyon aperture in line with sensor or license plate (holder) and parts that act as a “resonant box” (see Figure 10). Therefore, special considerations shall be given if the sensors are placed next to the mentioned parts. The action to avoid the canyoning effect must be agreed with Continental in an early stage of development. Possible actions are to place the sensors outside the main reflection directions that are marked with red color in Figure 10 or to move the sensor in front to have the canyoning structure behind the sensor surface.

If a placement outside the red areas is not feasible due to bumper design constraints, it is also possible to place the sensors inside the red areas and to mask the emerging ghost echoes by adaption of the sensor sensitivity parameters or mechanics. Specific verification tests need to be performed with the final vehicle bumper design, equipped with the final variable part setup, to evaluate and ensure sensor system functionality is not be affected by placement in these areas.

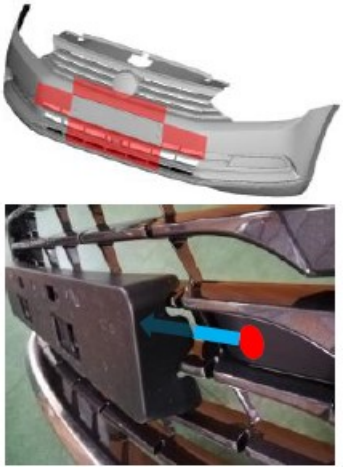


Figure 10: Main reflection directions of license plate (marked with red color in top figure) and canyoning effect

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

3.2.3 Special Consideration to the placement of sensors next to Protuberances

Protuberances, e.g. tow bar, spoiler, grill, can cause reflections of the transmitted ultrasonic wave or can modify the ultrasonic beam. Therefore, special considerations shall be given if the sensors are placed next to the mentioned parts. The action to avoid reflections/modified beams must be agreed with Continental in an early stage of development. A possible action is to move the sensor in front to have the protuberances behind the sensor surface.

If a placement next to protuberances cannot be avoided due to bumper design constraints, it is also possible to place the sensors next to the protuberances and to mask the emerging ghost echoes by adaption of the sensor sensitivity parameters or mechanics. Specific verification tests need to be performed with the final vehicle bumper design, equipped with the final variable part setup, to evaluate and ensure sensor system functionality is not be affected by placement in these areas.

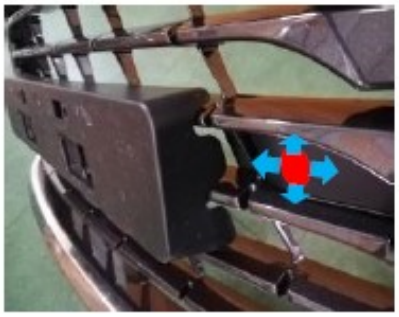


Figure 11: Reflections caused by Protuberances

3.2.4 Avoidance of sensor placement within or directly next to air turbulence generation systems

As air turbulences disturb the ultrasonic wave characteristics the sensors should not be placed within or directly next to air turbulence generation systems as e.g. exhaust system, cooler fan.

3.2.5 Special considerations for metal bumper

Special considerations shall be done if the sensors are placed at a metal bumper regarding the mechanical decoupling of the sensor and its holder from the bumper.

If above guidelines cannot be followed, exchange Computer Aided Design (CAD) data with Continental in earliest state for further discussion.

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

3.3 Sensor Painting

The sensors can be painted. Guidelines for the painting will be provided separately.

4 Decoupling Ring

The decoupling ring is a mandatory part of the system that ensures the mechanical decoupling between the sensor membrane and the sensor holder, housing and vehicle bumper. The decoupling ring is flush mounted with the sensor membrane.

5 Bracket

The sensor must be mounted to the bumper with a sensor bracket. The bracket can either be provided by Continental or by the customer.

If the bracket is provided by the customer, it is recommended to use the in the following described design. If another design is used, the bracket and fixation method with the bumper is recommended to be validated by the customer and agreed with Continental.

During installation the axial forces maintained from outside on the membrane shall not exceed 150N and the axial forces to the sensor housing shall not exceed 25 N.

Continental proposes to use the bracket design according to the AK2 standard [1]. The bracket is glued at the bumper with the sensor either fixed by through mounting or without through mounting (1 joint solution).

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

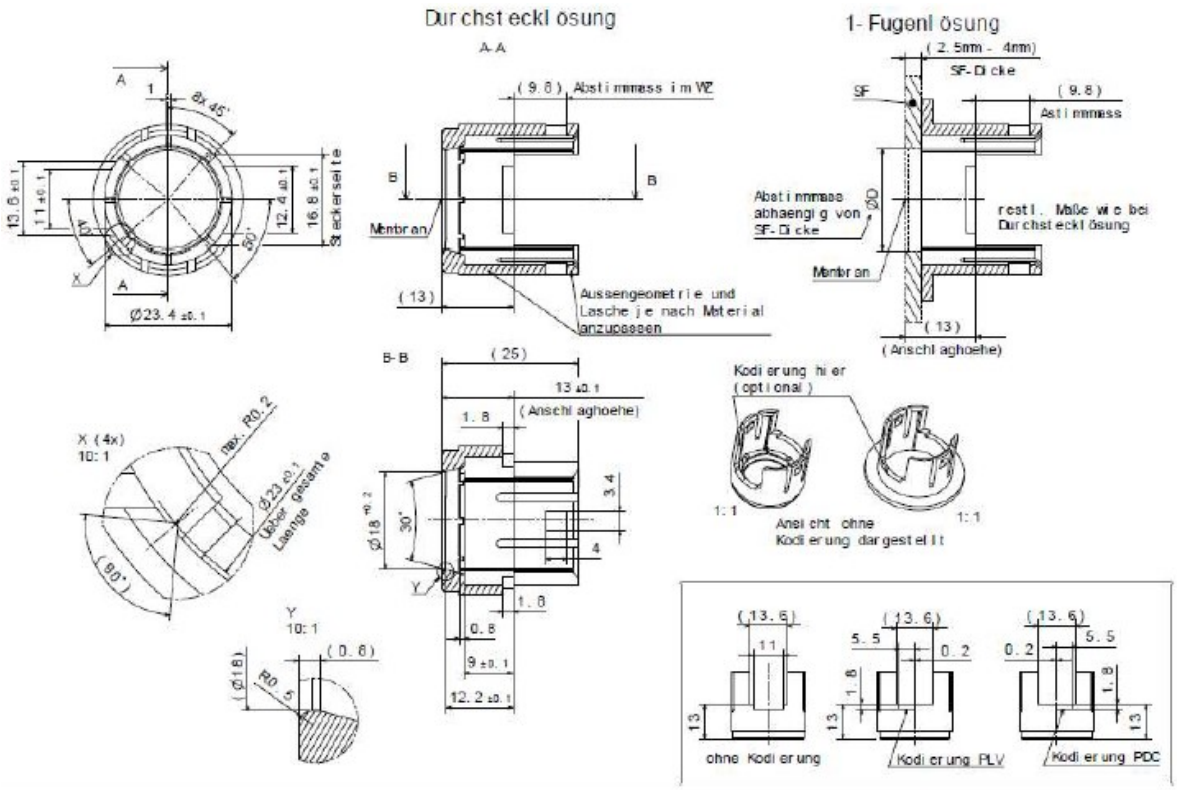


Figure 12: Sensor holder core model in glued version with (left) and without (right) through mounting (1 joint solution, final dimensions part of Mechanical drawings)

6 Heat Dissipation

At the mounting position the specified maximum ambient temperature of 95°C shall not be exceeded. In case that the maximum internal temperature is exceeded, the control unit will switch the sensor to protection mode.

7 Weight

The weight of the sensor unit is less than 15g without bracket and decoupling ring.

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

8 Connector Orientation

Each sensor can be placed on each mounting position at the bumper, connector facing to the left or to the right for the front/rear sensors and connector facing to the front/rear for the side sensors relative to vehicle x-axis.

9 Mounting Optimization

Continental provides support to analyze and optimize the mounting situation at every stage of the development of the vehicle.

Therefore, several measurement and simulation facilities are available:

- Simulation
- Vehicle FoV Test (ISO 17386)
- Ground Immunity Adaption
- Data capturing based on test driving

9.1 Simulation

The Field-of-View coverage simulation / optimization approach permits to save time, spare parts and time expensive tests, for the application tuning activities and in terms of deriving the best practicable sensor position setup.

Depending on several input configurations and boundary conditions an optimized setup shall be derived to fulfill predefined restrictions.

This is useful especially at the beginning of the vehicle definition where design could easily change and bumper/vehicle aren't available yet.

9.1.1 Solution Concept

The simulation solution will be conducted within 3 stages

1. **First stage: Optimization**

   Determine several local minima via simplified, fast calculating models based on predefined KPIs, boundary conditions & restrictions.

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

2. **Second stage: Verification**

   Run determined local minima via high resolution raytracing environment to verify global minimum as a practicable sensor position setup.

3. **Third stage: Validation**

   Run global minima in dynamic scenario testing environment using ground truth and/or raytracing USS models in terms of validation concerns.

9.1.1.1 First Stage: Optimization

Based on prescribed inputs, like boundary conditions and mathematical descriptions / evaluation functions, the optimization will find an optimum sensor positioning setup.

General approach is to feed the solution find necessary inputs and let the simulation converge towards a local minimum of the given environment (see Figure 13).

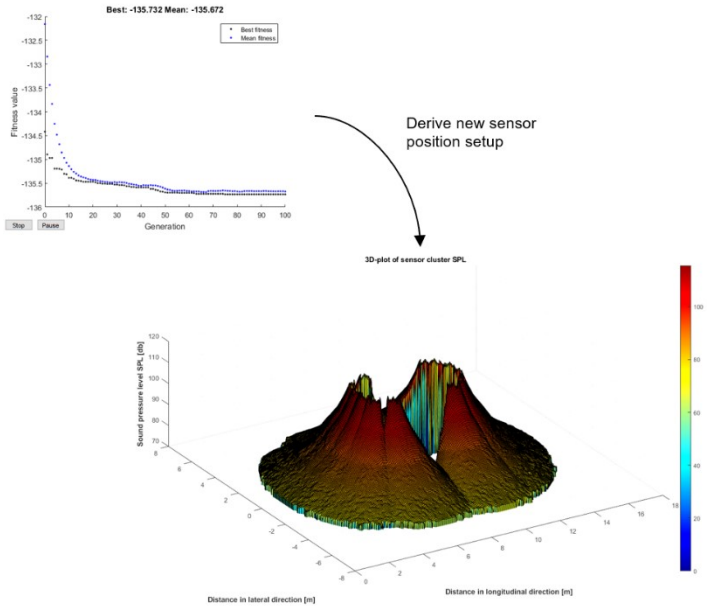


Figure 13: General simulation solution approach

The outcomes of the coverage simulator shall be confirmed with real test once the final design of the vehicle is available.

However, there are two basic problems considering a common FoV optimization. Both problems will be considered in a proper fidelity level. This will also be discussed with the customer himself to provide the best suitable solution based on individual needs.

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

The two common problems to consider are:

1. **Too many degrees of freedom** (in general neglect roll angles of the sensor rx)
    a. Displacement in sensor vision direction: **dx**
    b. Displacement in an orthogonal way corresponding to the senor vision direction: **dy**
    c. Displacement in sensors mounting height: **dz**
    d. Rotation around horizontal axis (pitch): **ry**
    e. Rotation around vertical axis (yaw): **rz**

In terms of reducing the total amount of DOFs, only 3 full varying DOFs will be uses (**dx, dy, rz**). The remaining two DOFs (**dz, ry**) will be covered via an intelligent usage of predefined antenna templates placed at different points of interest.

This matter will be explained within chapter 9.1.2.4.

2. **Which KPIs should be tested?**

    The KPIs will be divided into two separate groups

- **Basic KPIs** based on
    - Trilateration principles
    - 2D FoV coverage
    - Accumulated sound pressure amplitude and/or level
- **Complex KPIs** based on
    - Weighted operation modes (e.g. Near Field Detection - NFD)
    - Weighted areas (e.g. blind spots, vehicle path)
    - Weighted basic KPIs

The KPIs will be defined by either the customer, the supplier or both during discussions.

If the customer e.g. decides that the supplier shall determine the KPIs, there will be clarification discussions where the supplier will make a proposal of a useful KPI setup applicated onto the customers current sensor setup. This KPI proposal will be maintained throughout the entire optimization process and adapted if necessary.

The used KPIs have a huge impact on the simulation / optimization results and must be defined with caution in terms of avoiding unusable sensor positioning setups.

Further information can be found in chapter **0**.

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

The results of the optimization will be a complete sensor positioning setup, describing the tested inputs (**dx,dy,rz**). Figure 14 shows a visualization of the optimization results for different KPIs, e.g. trilateration principle, FoV coverage area and accumulated SPL.

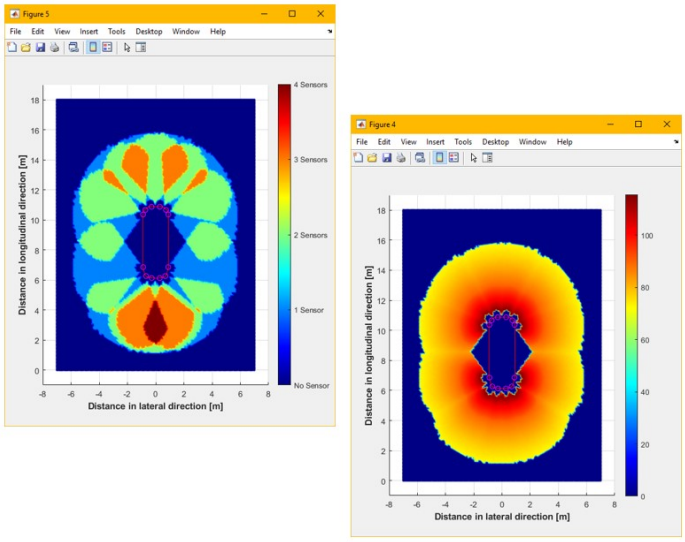


Figure 14: Possible visualization of optimization results

9.1.1.2 Second Stage: Verification

To clarify whether a derived sensor positioning setup by the optimization does fulfill its requirements, it will be verified in a high-resolution Raytracing environment.

This environment will be set up with

- usage of original CAD model in the simulation framework (if possible)
- virtual measurement based on AK2 grid measurement [1]
- tubes placed automatically via test automatization each 0.1m for capturing raytracing-based USS raw data
- take stage-1-optimization output as given sensor position setup to be tested

9.1.1.3 Third Stage: Validation

In terms of validating the verified results (sensor positioning setup) a dynamic scenario testing will be conducted.

Usual test cases are

- different traffic situations

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

- cut-in / cut-out scenarios
- overtaking situations
- pedestrians crossing

There will be a discussion with the customer about the exact test case catalogue.

9.1.2 Optimization Setup & Workflow

The clarification of the inputs and outputs of the optimization in combination with its parametrization represents the most important part of the simulation solution. As shown in Figure 15 there are 4 INPUTS to be defined. All of them are obligatory and cannot be skipped/neglected.

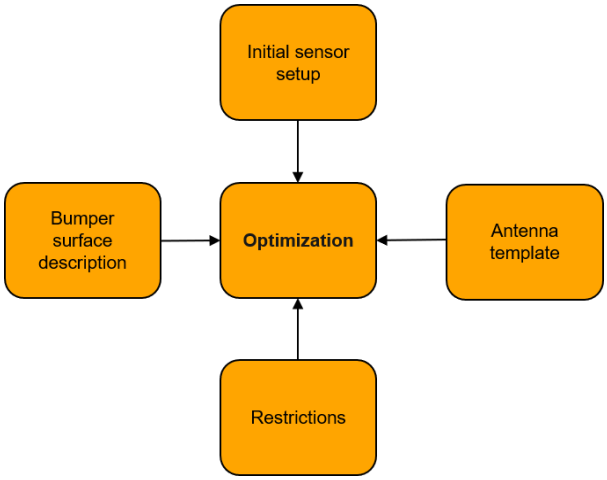


Figure 15: Necessary input information to the optimization solution

9.1.2.1 Initial sensor setup

As a starting point for the optimization an initial sensor setup is needed. This setup must have the minimum information for defining all sensors relative to the bumper. Necessary information is

- **dx, dy, rz**
- **dz**, in terms of defining the height of the bumper description (see chapter **9.1.2.2**)
- all above parameters either
    - for each sensor

    or
    - for half the amount of sensors if the sensors are located symmetrically identical (see chapter **9.1.2.3**)

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

The initial sensor setup can be

- described by the customer
- derived by the supplier throughout the CAD model
- estimated by the supplier based on bumper description (see chapter **9.1.2.2**)

9.1.2.2 Bumper surface description

In order of optimizing the sensors position right on the surface of the bumper, the solution, however, needs to be fed with a mathematical description of the bumper itself.

This mathematical description can be a

- 2D polynomial for a particular height
- 3D polynomial
- Point cloud

If these kinds of mathematical descriptions aren't available, the supplier has the possibility to derive via 3D CAD model all necessary information (see Figure 16).

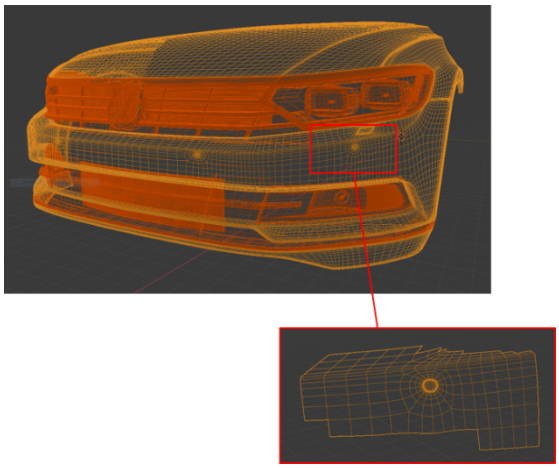


Figure 16: CAD cut-out of the optimization range around sensor 2

Accordingly, the customer should exchange his 3D CAD models in the needed format (e.g. obj-format) with the supplier.

9.1.2.3 Restrictions

Due to a more effective way of optimizing the sensor positions, the simulation must be fed with additional information concerning boundary conditions and restrictions. These shall help the solution to find the

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

optimum setup within a predefined, usable and practicable environment which may also enhance the calculations and simulations speed.

Possible restrictions can be

- Symmetric optimization

  e.g. left side sensor positions can be mirrored to the right side
- Predefined sensors rotation freedom

  e.g. +/- 20 deg or +10 / -20 deg etc.
- Predefined sensor position freedom

  Where are the optimization boundaries to the left and the right on the bumper?
- Sensor simulation range
- Customer defined restrictions

9.1.2.4 Antenna templates

Reducing the max DOFs to **dx,dy,rz** and neglecting the DOFs **dz,ry** (mounting height and sensor pitch) allows a faster calculation and more effective way to parametrize the solution environment. Experience has shown, that the neglected DOFs don't have such a big impact on the optimization results.

However, in terms of covering also the previously neglected missing DOFs **dz,ry** premeasured antenna templates will be used.

Therefore a measurement template for each mounting height and pitch, which are meant to be optimized, are necessary and will be provided throughout the supplier (see Figure 17).

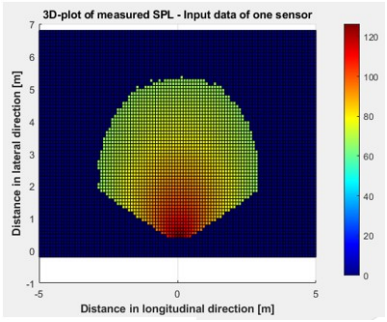


Figure 17: Example antenna template for one particular height and pitch

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

9.2 Vehicle FoV Test (ISO 17386)

The vehicle field of view test is conducted after the simulation when a first bumper design is available. The output of the coverage simulator and the placement of the sensors at the vehicle bumper can be verified by measuring the static detection field of view of the ultrasonic sensor system. In this test norm objects are placed in a 10 cm raster in the whole field of view of the vehicle bumper. For each new object position the detection of the sensors are evaluated. In this way restrictions and gaps in the field of view due to the application of the sensors can be detected.

9.3 Ground Immunity Adaption

With the final car platform test drives on different grounds to adapt the sensors immunity to ground reflections needs to be conducted. Sensor measurements on different grounds are collected to determine the different ground reflections that can be filtered in the sensor by adapting the sensor tuning parameter.

A list of different ground floors shall be agreed with the customer (e.g. pavement, asphalt, gravel).

10 Data capturing based on test driving

For final testing and releasing of a platform, test drives on the final car platform shall be conducted.

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.

11 References

[1] AK2 Standard: “Arbeitskreis Sensorik: Ultraschallsensorik für Einparkhilfesysteme 2.Generation “. Specification: 2.4.7-U-01a-2016_Gen2. Version 2.0, Change date: 10.10.2016

12 Table of Figure

Figure 1: Ultrasonic sensor with a 0° connector (left) and a 90° connector (right) 4

Figure 2: Field of view (FoV) of a single sensor 5

Figure 3: Detection Field of view (FOV) of a 12-sensor setup 6

Figure 4: Sensor naming and position 7

Figure 5: Mounting position and mounting angles 9

Figure 6: Sensor reference point (center of transducer) for α (alpha), β(beta), γ(gamma), Δy, h_z (Dimensions in mm, final dimensions part of Mechanical drawings) 9

Figure 7: Definition of angles: α, β, γ=0° 10

Figure 8: Pitch Angle β versus Mounting Height h_z for side sensors (US1, US6, US7, US12) 11

Figure 9: Pitch Angle β versus Mounting Height h_z for front/rear sensors (US2, US3, US4, US5, US8, US9, US10, US11) 11

Figure 10: Main reflection directions of license plate (marked with red color in top figure) and canyoning effect 14

Figure 11: Reflections caused by Protuberances 15

Figure 12: Sensor holder core model in glued version with (left) and without (right) through mounting (1 joint solution, final dimensions part of Mechanical drawings) 17

Figure 13: General simulation solution approach 19

Figure 14: Possible visualization of optimization results 21

Figure 15: Necessary input information to the optimization solution 22

Figure 16: CAD cut-out of the optimization range around sensor 2 23

Figure 17: Example antenna template for one particular height and pitch 24

Transmittal, reproduction, dissemination and/or editing of this document as well as utilization of its contents and communication thereof to others without express authorization are prohibited. Offenders will be held liable for payment of damages. All rights created by patent grant or registration of a utility model or design patent are reserved.